



Society of Petroleum Engineers

**SPE-229595-MS**

## **Stratigraphic and Sedimentological Characterization of An Unconventional Carbonate Deposit: An Example from the Late Cretaceous Shilaif Formation, United Arab Emirates**

Greg M. Baniak, Claire M. Woods, and Christian Abaco, PETRONAS Canada Energy, Calgary, Alberta, Canada; Nathan E. Leonard, Dana B.K. Kinnaird, Walid Kholeif, and Fatimah Rosly, PETRONAS Abu Dhabi, Abu Dhabi, United Arab Emirates

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229595-MS](https://doi.org/10.2118/229595-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

### **Abstract**

The Late Cretaceous (Cenomanian) Shilaif Formation is an unconventional oil deposit located in the Emirate of Abu Dhabi in western United Arab Emirates (UAE). Presently, it is undergoing early-stage appraisal drilling and development. Due to the reservoir in its natural state having a very low permeability (well below 1mD), traditional primary recovery techniques cannot be used. Instead, extraction of the oil will occur via hydraulic fracking of the reservoir. To better optimize landing zone placement of horizontal wells, which in turn will help lead to improved design of the wells, operations require a thorough understanding of the depositional fabric and stratigraphic architecture of the reservoir.

The Shilaif Formation represents a deep-marine carbonate succession deposited within an intrashelf basin. Using Arabian Plate boundary nomenclature present in offsetting wells from various offsetting countries, three primary sequences can be delineated (from oldest to youngest): K120 (Lower), K130 (Middle), and K140 (Upper). Each of these three sequences can be further sub-divided into lowstand (LST), transgressive (TST), highstand (HST), and falling stage (FSST) systems tracts and their corresponding parasequence sets. At the core level, massive bedded and bioturbated organic-rich carbonate facies in each of the three sequences frequently correspond with TST and early stage HST successions. Thin-sections reveal significant planktonic foraminifera and calcispheres. Biostratigraphic analysis of core samples suggests much of the organic material is a by-product of phytoplankton (algal) blooms. Conversely, massive bedded and bioturbated non-organic carbonate core facies in each of the three sequences commonly correspond with FSST and LST successions. Thin-section analysis reveals a higher abundance of fossiliferous fragments thus indicating sediment transport from adjacent platform margins into the basin. It must be noted that different combinations of these organic and non-organic facies, however, can occur within various systems tracts and therefore underpins the need to do detailed core logging and subsequent correlation across the field. Finally, many key stratigraphic surfaces are punctuated by examples of *Glossifungites* ichnofacies and their recognition in core are of critical importance towards building the basin model and understanding water benches.

## Introduction

Global demand for oil and gas is anticipated to continue increasing for the foreseeable future (OPEC World Oil Outlook, 2025). To help meet demand, global hydrocarbon production will need to continue shifting from conventional reservoirs to intervals once deemed inaccessible. Among others, these include heavy oil reservoirs and low-permeability unconventional reservoirs. Unconventional reservoirs are unique in that they commonly have porosities below 8% and permeabilities within the nanodarcy range. Given the very low matrix permeability, economic hydrocarbon extraction is achieved through horizontal drilling and subsequent multistage hydraulic fracturing. By increasing the accessible surface area of the reservoir and creating a series of artificial (man-made) flow networks, international companies have been able to stimulate successions once considered "dead zones" into highly profitable intervals. As seen in many operations globally, companies have employed a manufactured mindset wherein well pads are drilled adjacent to each other to help cover the greatest surface area possible. A primary reason tight pad drilling can occur is because many unconventional reservoirs are very thick and continuous. As a result, many operators have incorrectly deemed these reservoirs as "homogenous".

Given the apparent consistency of the reservoir at the field-scale, in combination with extremely high net-to-gross values, many geologists and engineers have overlooked basic first principles when it comes to utilizing stratigraphic mapping to understand these unconventional systems. A large reason is due to the massive first glance appearance of the rock facies at the full diameter core level, which lends itself to erroneous interpretations that the reservoir is homogenous and therefore predictable. Within a conventional basin, subsurface teams will spend years, if not decades, identifying and characterizing elements related to the petroleum system. Among others, these include confined stratigraphic and structural traps identified on 3D seismic, depositional fairway trends observable in core, and porosity and permeability relationships observed on petrophysical well-logs. Conversely, defining elements of unconventional reservoirs is that they are pervasive throughout a large geographical area, do not occur in discrete structural or stratigraphic traps, and are often self-sourced. For instance, the Triassic Montney Formation shale resource play in western Canada is approximately equivalent to the size of England. Perhaps most importantly, dry-wells (i.e., zero net pay) can occur within a conventional basin and large amounts of up-front capital are spent on appraisal programs understanding a confined accumulation of hydrocarbons before final investment decisions can occur. On the other hand, within unconventional reservoirs, the existence of hydrocarbons is well known (i.e., zero net pay is nearly impossible) and rather the challenge centers around the commerciality of production and scale of development. [Burkholder et al. \(2012\)](#) showed that boundaries between exploration, appraisal, and development are often unclear for unconventional reservoirs. In short, the difficulty is not making a discovery but in commercially developing it.

Since 2010, most of the unconventional oil and gas growth has occurred within Argentina, Canada, and United States. For instance, contribution from shales for total United States hydrocarbon production has increased from 5% in 2007 to over 60% in 2023 ([McMahon et al., 2024](#)). Over the past decade, many middle Eastern countries have also begun exploring and appraising their own unconventional resources hoping to build on the successes seen in the western Hemisphere ([Gramajo and Rached, 2022](#) and references therein). Examples include oil from Bahrain (Khaliq Al Bahrain Basin), gas condensates from Saudi Arabia (Jafurah Basin), and both gas (Ruwais Diyab concession) and oil (Al Dhafra region) from United Arab Emirates. Of note, all three of these countries are targeting unconventional reservoirs that are carbonate-rich and are therefore different relative to the siliciclastic-rich unconventional examples present across Argentina, Canada, and United States. Given the inherent complexities present within carbonates (e.g., diagenesis) and their associated impacts to reservoir quality, developing a proper stratigraphic and depositional framework is even more critical within these unconventional carbonate systems. Herein, we present an example from southern United Arab Emirates (Late Cretaceous Shilaif Formation) that characterizes an unconventional

carbonate reservoir that uses core facies, biostratigraphy, thin-section petrography, and petrophysical well-logs as a foundation.

## Study Area

The UAE has been a steady producer of oil and natural gas for over the past 50 years, with production coming from high-quality conventional reservoirs. Within the Abu Dhabi National Oil Company (ADNOC), active efforts are currently being made to diversify production towards unconventional resources. To help achieve this goal, extensive efforts in recent years have occurred to understand organic-rich source rocks that could be targeted for unconventional extraction. In 2015, the United States published a summary of the Technically Recovery Shale Oil and Gas Resources for the UAE (EIA, 2015). The three intervals highlighted as prospective opportunities for unconventional upside include the Silurian Qusaiba shale, Jurassic Diyab carbonate, and Cretaceous Shilaif carbonate. Within the EIA report, it was noted that the Cenomanian Shilaif Formation has 367 billion barrels of risked oil in-place and 22 billion barrels of risked, technically recoverable.

Recognition of the Shilaif Formation as an attractive unconventional opportunity dates back almost 20 years (e.g., [Taher, 2010](#); [Steiner et al., 2015](#)). To build on this opportunity, strong efforts have been made by ADNOC to attract foreign expertise to help explore, appraise, and ultimately develop these unconventional Shilaif oil deposits. A great example includes PETRONAS Abu Dhabi, who inked an agreement in 2022 with ADNOC to explore and appraise Unconventional Onshore Block 1 (UCO1) covering an area of 2,000 km<sup>2</sup> (~500,000 acres) in the Ghurab Syncline ([Fig. 1](#)). Building on this momentum, PETRONAS Abu Dhabi added adjacent Unconventional Onshore Block 5 (UCO5) to its portfolio in 2024, which covers an area of 4,000 km<sup>2</sup> (~1,000,000 acres). More recently, EOG Resources in 2025 was awarded Unconventional Onshore Block 3 (UCO3), covering an area of 3,609 km<sup>2</sup> (~900,000 acres). These recent agreements underscore the substantial resource potential of the Shilaif Formation and opportunities for global expansion of unconventional oil production.



Figure 1—Location map of the UAE and corresponding PETRONAS Abu Dhabi onshore oil blocks UCO1 and UCO5 in the Emirate of Abu Dhabi.

## Regional Context

The Shilaif Formation is Cenomanian in age and is part of the Wasia Group (Alsharhan and Nairn, 1988). Its lateral in-country geographical equivalent includes the Mishrif Formation, an interval characterized by the frequent occurrence of shelfal rudist buildups and grainstone shoals. The Mishrif Formation was deposited on the more proximal edges of the massive intrashelf basin present during the Cenomanian (Burchette, 1993; Van Laer et al., 2019; Bromhead et al., 2022). The Shilaif Formation itself was deposited in the distal portions of the intrashelf basin (Alsharhan, 1989; Vahrenkamp et al., 2015; Steiner et al., 2015). The organic content ranges between 0.1 to 9%, with some intervals having values of up to 15% (Taher, 2010; Steiner et al., 2015). Given these high organic carbon values (TOC), the Shilaif is believed to be a source rock for some of the major UAE Mishrif and Simsima reservoirs across the UAE (Gramajo and Rached, 2022; Alharthi et al., 2023).

## Methods

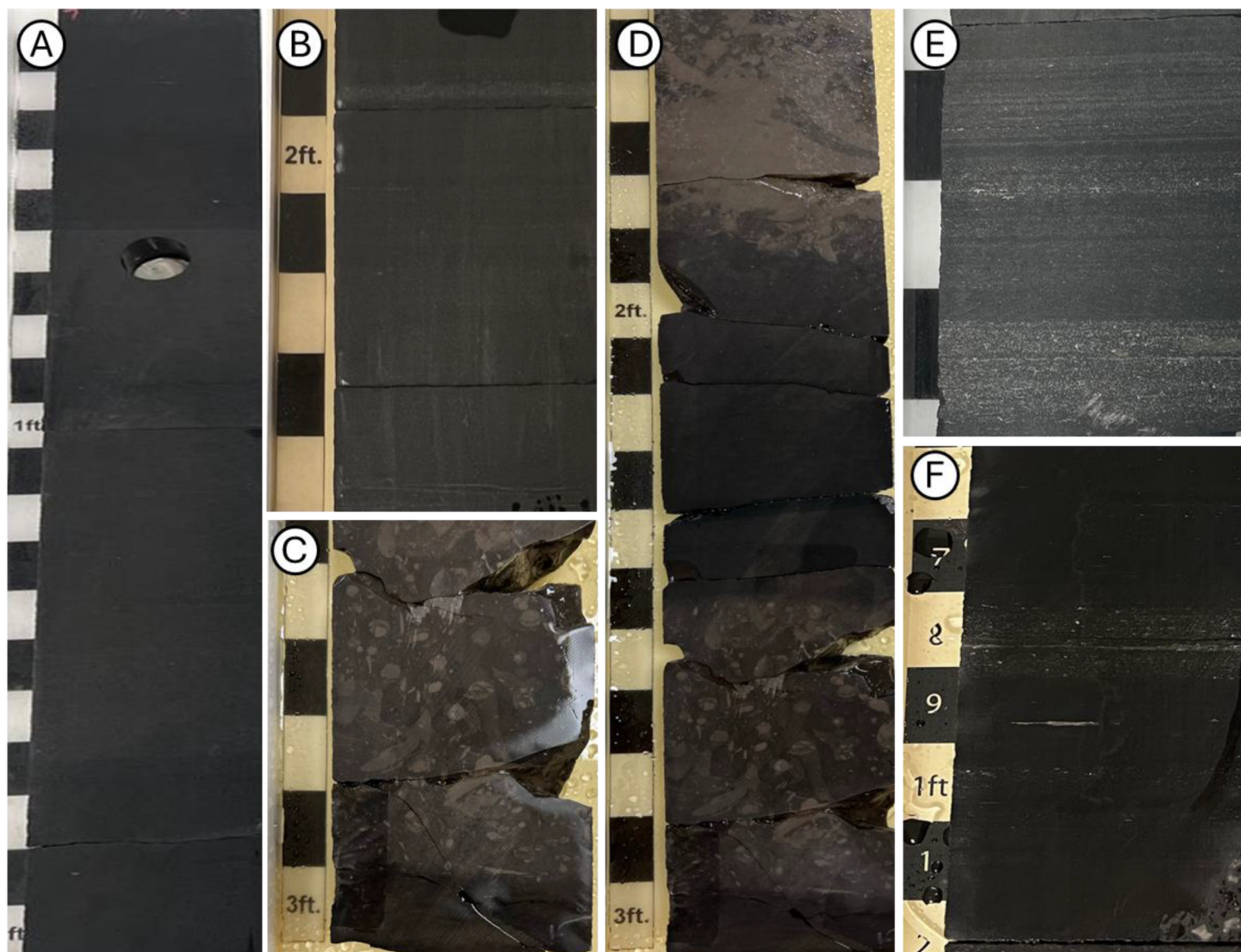
Using slabbled core samples recovered from three subsurface wells in UCO1, sedimentological and ichnological properties were assessed with the goal of identifying repeatable facies. The core facies were coded into excel and subsequently loaded into PETREL software for visualization and correlation purposes. Thin-sections were used to evaluate allochems present and biostratigraphic analysis was completed to better refine paleoenvironmental interpretations and aid in kerogen analysis of the organic matter present. NEFTEx Arabian Plate boundary nomenclature (e.g., K140SB) of Sharland et al. (2001) was applied to help correlate time-equivalent stratigraphic packages across the Shilaif Formation on well-logs. Important stratigraphic surfaces were identified based on core evaluation and referenced back to well-logs.



## Results: Stratigraphy and Associated Core Facies

The Shilaif Formation, and its lateral equivalent Mishrif Formation, form part of an expansive carbonate platform that stretched across much of the middle East during the Late Cretaceous ([Alsharhan, 1989](#); [Sharland et al., 2001](#); [Bromhead et al., 2022](#)). Because the intrashelf basin was long-lived across most of the Cenomanian, three primary stacked sequences ultimately formed within the Shilaif Formation and are related to the K120, K130 and K140 sequences ([Bromhead et al., 2022](#)). Sequence 1 (K120) can be subdivided into five parasequence sets, sequence 2 (K130) into four parasequence sets, and sequence 3 (K140) into six parasequence sets. The depositional sequences defined by [Hunt and Tucker \(1992\)](#) was utilized to subdivide each of the three sequences and corresponding parasequence sets into definable systems tracts. In short, this included usage of a lowstand, transgressive, highstand, and falling stage systems tracts (LST, TST, HST, and FSST respectively). Notably, ADNOC geologists (e.g., [Alharthi et al., 2023](#)) have also used a similar stratigraphic framework for the Shilaif Formation.

Six primary core facies can be classified and ultimately can be characterized as either abundant in organic carbon (TOC; [Figure 2](#)) or relatively clean (no TOC present; [Figure 4](#)). These facies include Massive TOC (Facies 1), Bioturbated TOC (Facies 2), Coarse-grained TOC (Facies 3), Massive Appearing, non-TOC (Facies 4), Bioturbated, Non-TOC (Facies 5), and Massive Appearing, non-TOC with Argillaceous Material (Facies 6). With relation to the stratigraphic framework, Facies 1 through 3 bear TOC and commonly correspond with TST and HST systems tracts. Facies 4 through 6 are commonly non-TOC bearing and correspond with FSST and LST systems tracts. These aforementioned groupings are not exact though, as different combinations of facies can occur in each parasequence set and corresponding systems tract. For instance, Facies 4 often can occur within an HST and vary in abundance from one location to another thus necessitating the need for detailed well-log correlation and facies descriptions.



**Figure 2—Examples of TOC-rich facies present within the Shilaif Formation. (a-b) Examples of Facies 1, exhibiting massive bedding within core. (c) Example of Facies 2, with high bioturbation intensities, sharply overlying Facies 1. (d) Example of alternating Facies 1 and Facies 2 over a ~1 m (3 ft) interval. (e-f) Examples of Facies 3, showing sharp based coarse-grained clasts within a predominately organic-rich Facies 1.**

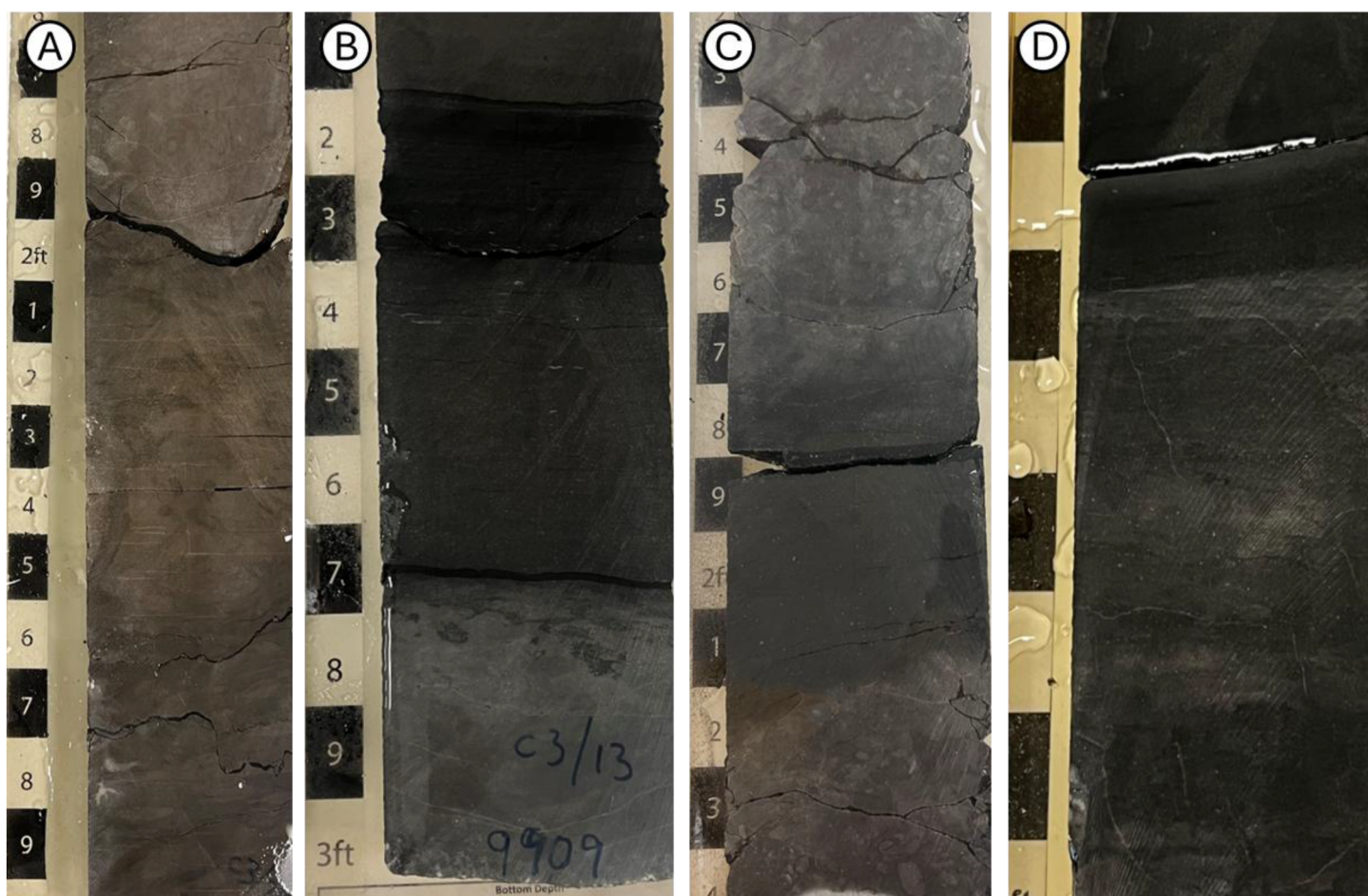


Figure 4—Examples of non-TOC facies within the Shilaif core. (a) Example of a fairly homogeneous appearing non-TOC limestone from Facies 4. Interbedded within are examples of solution seams related to Facies 6. (b) Facies 4 at the base of the core photo, transitioning into an example of Facies 3 and ultimately Facies 1. (c) Example of a bioturbated Facies 5 exhibiting sharp contacts with Facies 4. (d) Facies 6 with wispy concentrations of argillaceous material and low amounts of bioturbation.

### Massive TOC (Facies 1)

Facies 1 is a succession of dark, organic-rich mudstones to packstones (Fig. 2A-B). Sedimentary structures are often limited to planar laminations that will commonly transition into massive appearing structures. Texturally, the appearance of rock fabric at core level is similar to other unconventional rock fabrics seen globally (i.e., massive appearing and structureless). Most of the samples analyzed for palynofacies plot in the IX field of the ternary plot (after Tyson, 1995), indicating a suboxic to anoxic basin. Additionally, many of the palynofacies slides are dominated by amorphous organic material, which has mainly diffuse edges thus indicating a likely algal origin (i.e., phytoplankton blooms). Calcspheres and planktonic (*Globigerina*) foraminifera are superabundant as well, further supporting deposition within a restricted basinal setting (Fig. 3A-B).



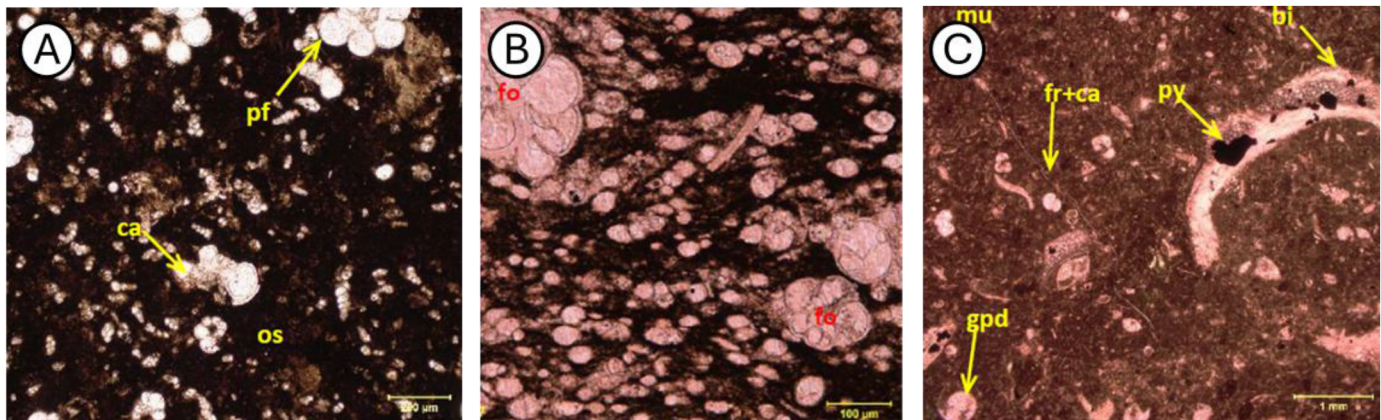


Figure 3—Thin-section photomicrographs. (a) Planktonic foraminiferal wackestone. Large foraminifera (pf) exist within the sample, along with secondary calcite (ca) replacement. The micritic matrix contains fine disseminated organic matter and organic streaks (os). (b) Planktonic foraminiferal packstone. Large foraminifera (fo) exist, with moderate amounts of calcareous shell fragments. Fine disseminated organic matter exists within the micritic matrix. (c) Skeletal packstone. Skeletal allochems present within the micritic matrix (mu). Examples include gastropods (gpd), rare echinoderms and bivalve fragments (bi). Fractures filled with calcite (fr+ca) also exist, along with pyrite (py).

### Bioturbated TOC (Facies 2)

Facies 2 is represented by a succession of highly bioturbated wackestones to packstones that often have sharp contacts with Facies 1 (Fig. 2C-D). Sedimentary structures are quite rare as the burrows occupy between 50 to 100% of the total volume. Although difficult to discern in many successions, the burrows are most comparable to examples of *Chondrites*, *Zoophycos*, *Planolites*, and rare *Thalassinoides* and reflect an example of a distal *Cruziana* ichnofacies (MacEachern et al., 2007). High bioturbation intensities, combined with low trace fossil diversities and elevated organic carbon, indicate a setting experiencing broad oxygen fluctuations (Bromley and Ekdale, 1984). Thickness variations and associated variations in trace fossil diversity within Facies 2, and their relationship with Facies 1, lend evidence towards fluctuations in oxygen content across the intrashelf basin during deposition. For instance, during late stages of LST and early stages of TST, Facies 2 often occurs in higher abundances relative to Facies 1 (i.e., 70:30 ratio) and has higher trace fossil diversity. As sea-level transgressed throughout the TST, laminated intervals of Facies 1 became more abundant and Facies 2 became thinner, more monospecific, and more infrequent thus suggesting increases in anoxic conditions.

### Coarse-Grained TOC (Facies 3)

Facies 3 in many regards can be considered an accessory to Facies 1. It is composed of thin beds (10 to 20 cm most commonly) of bioclastic or coarse-grained clasts (Fig. 2E-F). The contact with Facies 1 is very sharp and its abundance varies across most TST and HST parasequence sets from rare to semi-frequent. Sediment-gravity flows (e.g., turbidity currents, debris flows, grain flows) are a likely mechanism for these clast deposits (Middleton and Hampton, 1973). The proximal equivalent Mishrif Formation is a likely source of these fragments. Further studies, in addition to more core recovery, will be required to help determine exact provenance of these deposits and their potential significance to reservoir quality.

### Massive Appearing, Non-TOC (Facies 4)

Transitioning into Facies 4 through 6, organic-rich material becomes absent altogether. A common feature of Facies 4 in core is its homogeneous appearance due to the absence of any visible sedimentary structures or diagenetic overprints (Fig. 4A-B). Petrographically, it is a packstone to grainstone composed of planktonic foraminifera and calcispheres that also includes examples of thin-walled bioclastic debris (Fig. 3C). Higher-energy elements such as pelecypod valves and fragments, echinoderm fragments, gastropods, and crinoid fragments are examples of bioclastic debris and vary in abundance both vertically and laterally across the

Shilaif Formation. As seen in Facies 3, the bioclastic debris in Facies 4 likely originated on basin margin platforms and were transported to the basin as sediment gravity flows (Bromhead et al., 2022).

### **Bioturbated, Non-TOC (Facies 5)**

Similar to Facies 2, bioturbation within Facies 5 is also quite pervasive and can occupy up to 100% of the cored interval (Fig. 4C). Its presence is most common in FSST and LST successions, along with late stage HST parasequences in select wells. Trace fossil diversity (addition of examples such as *Teichichnus*) and burrow sizes (e.g., *Thalassinoides*) are greater than observed within Facies 2. Additionally, Facies 5 can commonly exceed 1m in thickness across many FSST parasequences. These overall increases reflect improving bottom water oxygenation, in addition to favorable food supply and sedimentation rates, and thus reflects a more proximal *Cruziana* ichnofacies (MacEachern et al., 2007).

### **Massive Appearing, Non-TOC with Argillaceous Material (Facies 6)**

Within Facies 6, the core appears non-homogenous at eye level and contains numerous examples of diagenetic alterations. Examples include solution seams, which often range from 1 to 10cm in continuous lateral thickness and can have TOC up to 20% in concentrations (Van Laer et al., 2019). Additionally, high amplitude stylolites are also observable. Each of these examples are likely a result of burial and subsequent diagenesis. Additionally, examples of wispy concentrations of argillaceous materials are common (Fig. 4D) and can be found dispersed throughout the matrix. A likely source of this clay material is biogenic aggregates and fecal pellets.

## **Results**

### **Global Anoxia and Preservation of Organic Facies During the Cretaceous**

Organic matter preservation within Cenomanian deposits exists across vast areas of the middle East and elsewhere globally. To help explain elevated organic matter production and subsequent preservation, one needs to look at the climatic conditions present. As a whole, the Cretaceous Period climate was significantly different from current modern levels. For instance, evidence exists showing the warmest ever reported temperatures during the early Late Cretaceous hot greenhouse. During these times, surface waters were greater than 35 °C and deep-ocean water greater than 20 °C (e.g., Clarke and Jenkyns, 1999; Forster et al., 2007; O'Brien et al., 2017). The Early to Late Cretaceous boundary (Albian to Cenomanian) is also an important marker globally as it reflects the beginning of a major eustatically controlled expansion of the Cretaceous seas (Haq et al., 1987, Haq, 2014). Sea-levels during the early Cenomanian were 215m higher than present day and near the end of the Cenomanian had reached up to 250m higher (Haq, 2014), representing the highest sea-levels recorded during the Cretaceous Period. Elevated atmospheric CO<sub>2</sub> levels (e.g., Barron et al., 1995; Wang et al., 2014) also existed for extended periods of time during the Late Cretaceous.

As a result of elevated sea-levels, increased surface temperatures, and increased CO<sub>2</sub> during the Cenomanian, conditions became ripe for elevated periods of marine anoxia. Ocean anoxic events (OAEs) have been recognized in the rock record to demarcate discrete intervals during which low-oxygen zones in the global ocean expanded significantly (Schlanger and Jenkyns, 1976; Jenkyns, 1980). These events represent prolonged and intense perturbations of the global carbon cycle (Arthur et al., 1987) and are therefore associated with widespread anoxia in most oceanic basins. The most widespread and best-defined example globally of a OEA is believed to be the late Cenomanian to early Turonian (defined as OAE2), within which carbon-rich sediments were concentrated (Jenkyns, 1980). Additionally, a mid-Cenomanian OAE (Coccioni and Galeotti, 2003) appears to represent a less intense precursor event of the Cenomanian-Turonian OAE2. Many different hypotheses exist to explain the processes triggering and maintaining global anoxia and enhanced primary productivity during various Cretaceous OAEs. Among others, these include



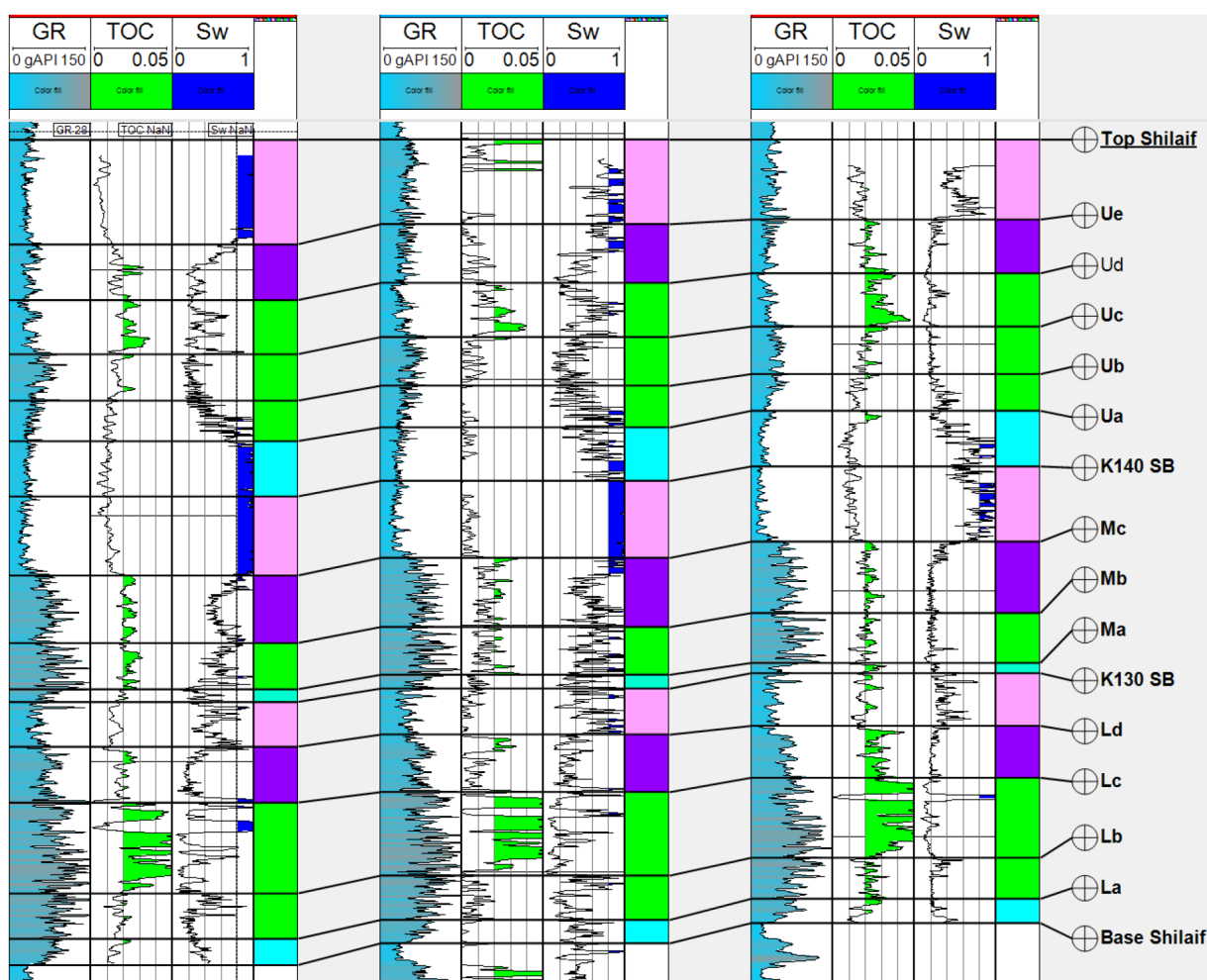
nutrient input into the ocean system due to activity associated with large igneous provinces (Trabucho-Alexandre et al., 2010) and nutrients from continental areas mobilized via major marine transgressions (Mort et al., 2008).

Within Facies 1 through 3, a significant source of the organic material present is believed to have been derived from phytoplankton blooms and its associated by-products. To help explain the prominence of phytoplankton during the OEA2, one can look at modern examples. In modern marine settings, studies have shown that some marine phytoplankton are observed to show significant stimulation of growth due to enhanced atmospheric CO<sub>2</sub> (Beardall et al., 2009). Additionally, marine snow is ubiquitous (Alldredge and Silver, 1988) in modern marine settings and provides organic detritus into deeper marine settings. Marine snow is most abundant below regions of enhanced primary production, particularly in association with phytoplankton blooms (Turner, 2002, 2015 and references therein).

### Facies Variability

Within all unconventional basins, a defining element is the apparent "homogeneity" of the unconventional reservoir. Core facies and their associated well-log signatures appear at first glance to be very thick, vertically continuous, and contain no obvious stratigraphic boundaries. Perhaps more importantly, many of these elements appear in subsurface holes long distances apart from one another (i.e., greater than 10km) thus leading to the belief that the reservoir is predictable and not in need of detailed subsurface analysis. Over the past decade, excellent case studies have emerged from western Hemisphere basins demonstrating how ichnology, sedimentology, and stratigraphy can be applied to characterize unconventional siliciclastic reservoirs (e.g., Kietzmann et al., 2014; Wilson et al., 2020; Baniak et al., 2023). These technical studies have highlighted how reservoir facies are in fact quite variable and contain elements which will have a distinct influence on petrophysical and geomechanical properties. As a result, production performance from one pad location to another has been observed to be quite variable and the identification of favorable reservoir facies has emerged as a leading indicator of well success. An additional element, although not addressed in this paper directly, will also likely include the influence of diagenesis on reservoir quality within the Shilaif carbonates and will form the basis for further research in the coming years.

The Shilaif Formation is unique in that each of the three sequences show a repeating succession of organic-rich facies interbedded with clean non-organic limestones. Given the critical importance of developing type-curve areas and identifying zones for landing zone placement within the appraisal phase of development, it is important to further subdivide each of these three sequences into parasequence sets and corresponding depositional sequences (i.e., LST, TST, HST, and FSST: Fig. 5). By doing so, core facies trends and associated petrophysical properties can be better understood over a larger geographical area which in turn will help better understand production trends in the future.



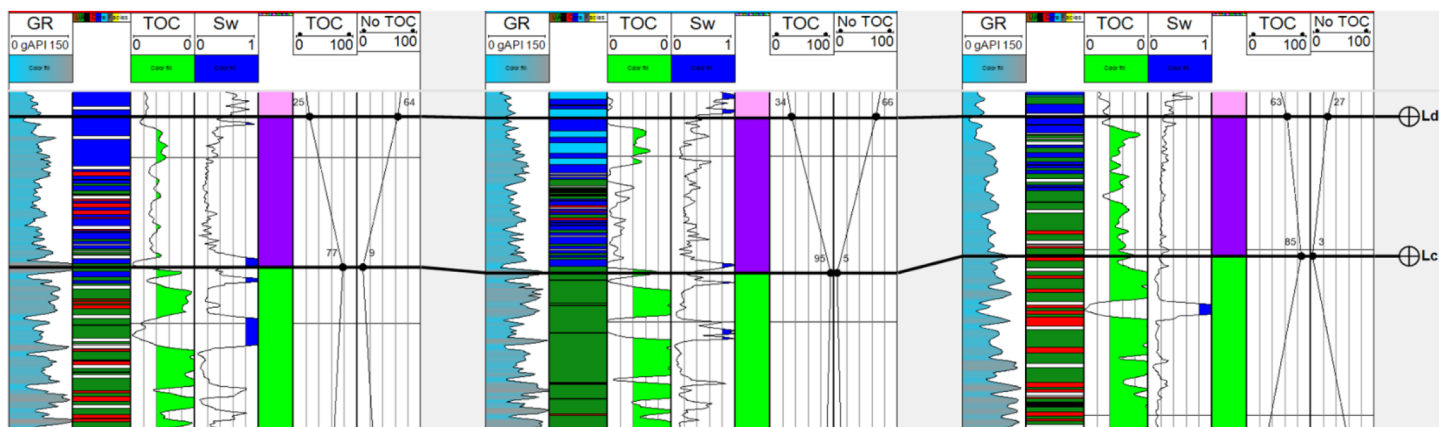
**Figure 5—A cross-section across UCO1 highlighting the three primary sequences within the Shilaif Formation. Sequence 1 can be subdivided into five parasequence sets (La to K130 SB), sequence 2 into four parasequence sets (Ma to K140 SB), and sequence 3 into six parasequence sets (Ua to Top Shilaif). The corresponding systems tracts within each parasequence set are color coded (far right column on each well) as follows: LST (light blue), TST (green), HST (purple), and FSST (pink). Each of the parasequence sets are correlated across UCO1 using Gamma Ray (GR), Organic Carbon (TOC), and Water Saturation (Sw). To help identify trends, a 2% cut-off was used for TOC and an 80% cut-off for Sw.**

In long-lived, multi-sequence intrashelf basins, such as the Shilaif, sea-level changes had a profound impact on depositional environments and corresponding accommodation space. Within each of the three sequences, the sequence boundary (SB) is placed at the correlative conformity that defines the base of the LST. On well-logs, the contact between the FSST and LST in the K130 and K140 sequence boundaries is nearly indistinguishable and requires core analysis along with biostratigraphy. A primary reason is due to the clean (non-TOC) nature of the sediments and hence relatively clean appearing nature of the gamma-ray (GR). Regional cross-section mapping by [Bromhead et al. \(2022\)](#) suggest these sequence boundary surfaces are of vast regional extent and can extend across much of the middle East.

The LST within each of the three sequences are quite thin and often composed of Facies 4 through 6. Due to shedding from platform margins, varying proportions of bioclastic debris can be found across UCO1. These debris deposits, and subsequent diagenetic alterations, will likely have impacts on porosity and permeability trends and must be the focus of future studies. Just as importantly, due to the reduced carbonate factory present during these lowered sea-level periods, and increased shedding from the platform margins, conditions for organic deposition and subsequent preservation were limited. The maximum regressive surface can be observable in select core examples as a thin fossiliferous lag deposit (e.g., K120 LST-TST boundary). During the early stages of the TST, the rate of sea-level rise increased and the first presence

of Facies 1 occurs. Notably, bioturbation via Facies 2 is often very intense and can be seen in numerous parasequence sets within the early stages of the TST. As transgression continued into the later stages of the TST, Facies 2 becomes less abundant and eventually dissipates completely. On other hand, Facies 1 increases in both abundance and overall thickness. The mixture of bottom-water anoxia, in combination with increased water temperatures and water stratification, was likely a major contributing factor to the elimination of deposit-feeding organisms.

The maximum flooding surface (MFS), which separates the TST and HST, is commonly difficult to discern. Within the Shilaif core, and corresponding well-log signature, the MFS was placed where it appears that a progradational stacking pattern starts forming again. A notable feature of this progradation was the fact that non-organic facies started being deposited again. Importantly, depending on geographic location, significant differences in Facies 1 and 3 abundances versus Facies 4 and 5 can be seen in different wells across UCO1 within the HST (Fig. 6). Leading into the FSST, the notable shallowing-upward trend from deeper-water non-TOC packstones and grainstones is observed in all cored wells. Bioturbation via Facies 5 is often prominent within the FSST and reflects the return of hospitable bottom water conditions.



**Figure 6—A cross-section of parasequence sets Lc and Ld showing the variation in facies across UCO1. The core facies are highlighted in the second track (green= Facies 1, black= Facies 2, red= Facies 3, dark blue= Facies 4, light blue = Facies 5, grey = Facies 6, white= preserved core). Facies 1 through 3 were then combined in one track (denoted TOC, scale 0-100%) and Facies 4 through 6 were combined in one tracked (denoted No TOC, scale 0-100%). Within the Lc (TST), the majority are facies are TOC-rich in all three wells. Transitioning into Ld (HST), a considerable change occurs. On the far right well, over 60% of the facies are TOC-rich and roughly 30% are non-TOC. As one transitions to the middle well, a complete reversal occurs as roughly 30% of the facies are TOC-rich and roughly 60% are non-TOC. A similar trend is seen in the far left well. Although the impact of these facies changes are unknown to production trends at the moment, they do reflect an import feature present within the Shilaif basin and will likely help form key geological type-curve area trends.**

## Recognition of Important Stratigraphic Surfaces in Core via Ichnology

Within the field of ichnology, three substrate-controlled ichnofacies have been long established and include *Glossifungites* (firmground), *Trypanites* (hardground), and *Teredolites* (woodground). Studies in the early 1990s highlighted the utility of recognizing firmground surfaces attributable to the *Glossifungites* ichnofacies towards delineating surfaces of sequence stratigraphic significance (e.g., Savrda, 1991; MacEachern et al., 1992). Fundamentally, *Glossifungites* ichnofacies are an assemblage of trace fossils characterized by robust, sharp-walled (locally scratch-marked), unlined, passively filled, vertical to sub-vertical domichnia, excavated into semi consolidated (firmground) substrates. Within the Shilaif Formation core, *Glossifungites* are a common feature and exist at select key stratigraphic surfaces in the FSST, SB, and LST depositional sequences. The identification of *Glossifungites* at the HST-FSST boundary carries extra significance and will form the basis of further discussion below.

Fundamentally, the contact between the HST and overlying FSST deposits are often erosional and display sharp transitions. As defined by Plint (1988), these sharp contacts reflect regressive surfaces of



marine erosion (RMSE). The RMSE is ideally suited for omission suite colonization as the discontinuity is cut in a subaqueous marine setting. The dewatered or firm substrate can become colonized again and ultimately burrowed prior to burial during the forced regressive progradation. *Glossifungites* surfaces can stretch across distances greater than five kilometers in outcrop (e.g., Jurassic strata from Saudi Arabian outcrops: [Eltom and Hasiotis, 2019](#)) and are therefore excellent stratigraphic markers at intervals such as RMSE. Another defining feature of *Glossifungites* surfaces is their high connectivity. [Eltom et al. \(2019\)](#) showed that 90% connectivity could be established at only 10% bioturbation intensity for examples of *Thalassinoides* within a *Glossifungites* ichnofacies. Furthermore, due to diagenetic overprints present within the burrows relative to the matrix, enhanced porosity and permeability is a common feature ([Baniak et al., 2022](#) and references therein). Excellent instances of enhanced reservoir permeability are common at select *Glossifungites* surfaces in Saudi Arabia (Upper Jurassic Arab-D Member, Ghawar Field; [Pemberton and Gingras, 2005](#)) and Qatar (Upper Jurassic Arab-D Member, Dukhan Field; [Trabelsi and Beg, 2000](#)).

In the Shilaif Formation, it has been recognized that significant water-bearing intervals are present ([Van Laer et al., 2019](#)). Among others, these water benches pose major challenges to horizontal well landing zone placement and fracture optimization. The most significant example is a water bench present within the K130 FSST and overlying K140 LST ([Fig. 5](#) of this study and as defined by [Alharthi et al., 2023](#)). At the core level, numerous examples of *Glossifungites* ichnofacies are observable within the K130 FSST and overlying K140 LST. Most notable, however, is a *Glossifungites* example observable at the K130 HST-FSST contact that can be traced in different wells greater than 15 km apart ([Fig. 7](#)). The select example also shows oil-stained burrows across an interval roughly 0.3m (1 ft) in thickness and contains elements similar to *Thalassinoides* and *Zoophycos*, suggesting it is likely a distal expression of the *Glossifungites* ichnofacies ([Burton and MacEachern et al., 1997](#)). Given the *Glossifungites* present at K130 HST-FSST formed pre-basin alteration, its distance across the basin is likely quite vast and will undoubtedly be confirmed with additional cores in the future. Proper diagenetic conditions also existed that enabled enhanced porosity and permeability to exist within the *Thalassinoides* and *Zoophycos* burrows relative to the surrounding reservoir facies. A defining feature of the Shilaif basin is the presence of a major anticline that separates the west syncline (Ghurab) from the east syncline (Falaha) ([Dasgupta et al., 2017](#); [Van Laer et al., 2019](#)). Given that oil is believed to have migrated short-term distances from TOC rock facies into the cleaner non-TOC facies ([Steiner et al., 2015](#); [Van Laer et al., 2019](#)), it is highly probable that oil encountered this highly permeable *Glossifungites* horizon and migrated from the syncline up-dip into the anticline, thus leaving many successions of the K130 FSST and overlying K140 LST as water-bearing.

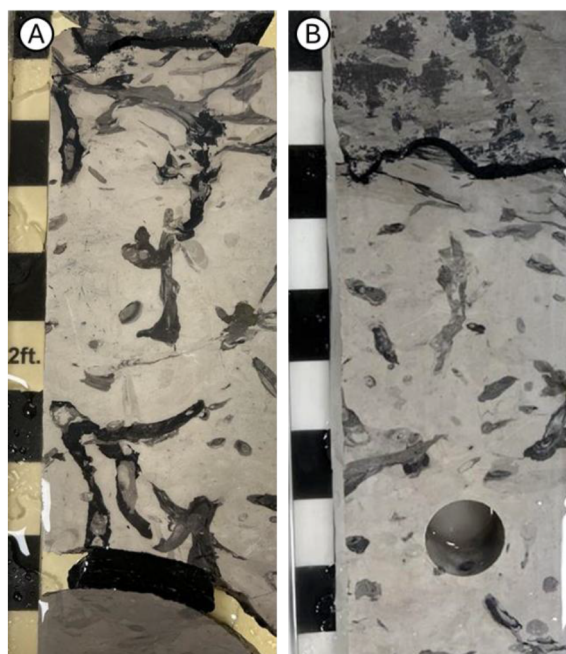


Figure 7—Identical appearing *Glossifungites* ichnofacies present in two wells greater than 15 km apart at the K130 HST-FSST boundary.

## Summary and Conclusions

The Shilaif Formation represents an emerging opportunity for expansion of unconventional oil production in the UAE. Within this study, we demonstrate the practical application of stratigraphic and sedimentological principles for assessing reservoir quality in an unconventional reservoir that is currently undergoing early-stage field-wide appraisal and development. In the case of the Shilaif Formation, primary reservoir facies are either abundant in organic carbon (TOC) or relatively clean (no TOC present). These facies, and their overall abundances, vary both vertically and laterally across block UCO1. To understand these facies variations, sub-division of the reservoir into mappable parasequence sets and depositional sequences (LST, TST, HST, FSST) is critical. Lastly, the demarcation of key stratigraphic surfaces via *Glossifungites* ichnofacies is critical towards underpinning the regional model and ultimately understanding challenges such as water benches present across the block.

## Acknowledgments

Many thanks to PETRONAS management in both Calgary and Abu Dhabi for their continual support and permission to publish this data. Additionally, we would like to thank ADNOC subsurface staff and management for their gracious demeanor and in-depth discussions surrounding the Shilaif Formation that helped crystallize many of the results presented herein. A large thank you to ADNOC and Stratum Reservoir core facility staff for locating core and letting us use their facilities for core observations. Finally, we would also like to thank staff at Future Geoscience for in-depth discussions regarding biostratigraphy and regional trends observed in core samples.

## References

- Alharthi, A., Baig, M., Van Laer, P., Al Marzooqi, H., Al Hashmi, A., Brooks, T., Al Bannay, A., Al Blooshi, A., and Al Braiki, M. 2023. Hidden success of unconventional oil play behind robust integration of subsurface G&G data. A case study of Shilaif unconventional oil play, Abu Dhabi, UAE. Presented at URTEC, Denver, Colorado, USA, 13-15 June. URTEC-3857395-MS. <https://doi.org/10.15530/urtec-2023-3857395>
- Allredge, A.L. and Silver, M.W. 1988. Characteristics, dynamics and significance of marine snow: *Progress in Oceanography* 20 (1): 41–82. [https://doi.org/10.1016/0079-6611\(88\)90053-5](https://doi.org/10.1016/0079-6611(88)90053-5)

- Alsharhan, A.S. and Nairn, A.E.M. 1988. A review of the Cretaceous formations in the Arabian Peninsula and Gulf: Part II. *Mid-Cretaceous (Wasia Group) stratigraphy and paleogeography: Journal of Petroleum Geology* **11** (1): 89–112. <https://doi.org/10.1111/j.1747-5457.1988.tb00803.x>
- Alsharhan, A.S. 1989. Petroleum Geology of the United Arab Emirates: *Journal of Petroleum Geology* **12** (3): 253–288. <https://doi.org/10.1111/j.1747-5457.1989.tb00197.x>
- Arthur, M.A., Schlanger, S.O., and Jenkyns, H.C. 1987. The Cenomanian-Turonian oceanic anoxic event, II. Palaeoceanographic controls on organic matter production and preservation. In: Brooks, J. and Fleet, A.J., eds., *Marine Petroleum Source Rocks: Geological Society, London, Special Publications* **26**: 401–420. <https://doi.org/10.1144/GSL.SP.1987.026.01.25>
- Baniak, G.M., La Croix, A.D., and Gingras, M.K. 2022. Recent advancements in characterizing permeability and porosity distributions in bioturbated flow media: *Earth-Science Reviews* **232** (101462): 22 pp. <https://doi.org/10.1016/j.earscirev.2022.104162>
- Baniak, G.M., Moslow, T.F., Michailides, S., and Adams, M.G. 2023. Sequence stratigraphic architecture of the Lower Triassic Montney Formation, northeastern British Columbia, Canada: *AAPG Bulletin* **107** (2): 283–310. <https://doi.org/10.1306/10252221113>
- Barron, E. J., Fawcett, P.J., Peterson, W.H., Pollard, D., and Thompson, S.L. 1995. A "simulation" of Mid-Cretaceous climate. *Paleoceanography and Paleoclimatology* **10** (5): 953–962. <https://doi.org/10.1029/95PA01624>
- Beardall, J., Stojkovic, S., and Larsen, S. 2009. Living in a high CO<sub>2</sub> world: impacts of global climate change on marine phytoplankton: *Plant Ecology & Diversity* **2** (2): 191–205. <https://doi.org/10.1080/17550870903271363>
- Bromhead, A.D., van Buchem, F.S.P., Simmons, M.D., and Davies, R.B. 2022. Sequence stratigraphy, palaeogeography and petroleum plays of the Cenomanian-Turonian succession of the Arabian plate: An updated synthesis: *Journal of Petroleum Geology* **45** (2): 119–161. <https://doi.org/10.1111/jpg.12810>
- Bromley, R.G. and Ekdale, A.A. 1984. Chondrites: A trace fossil indicator of anoxia in sediments: *Science* **224** (4651): 872–874. <https://doi.org/10.1126/science.224.4651.872>
- Burkholder, M.K., Coopersmith, E.M., and Schulze, J. 2012. Appraisal excellence in unconventional reservoirs. Presented at SPE Canadian Unconventional Resources Conference, Calgary, Alberta, Canada, 30 October–1 November. SPE-162776-MS. <https://doi.org/10.2118/162776-MS>
- Burton, J.A. and MacEachern, J.A. 1997. The distal expression of the Glossifungites ichnofacies: Implications for discontinuity recognition and the ichnofacies concept. In: Beauchamp, B., ed., *Sedimentary Events and Hydrocarbon Systems: CSPG–SEPM, Joint Convention*: 50 pp.
- Clarke, L.J. and Jenkyns, H.C. 1999. New oxygen-isotope evidence for long-term Cretaceous climate change in the Southern Hemisphere: *Geology* **27** (8): 699–702. [https://doi.org/10.1130/0091-7613\(1999\)027<0699:NOIEFL>2.3.CO;2](https://doi.org/10.1130/0091-7613(1999)027<0699:NOIEFL>2.3.CO;2)
- Coccioni, R. and Galeotti, S. 2003. The mid-Cenomanian event: prelude to OAE 2. *Palaeogeography, Palaeoclimatology, Palaeoecology* **190**: 427–440. [https://doi.org/10.1016/S0031-0182\(02\)00617-X](https://doi.org/10.1016/S0031-0182(02)00617-X)
- Dasgupta, S., Raina, I., Povstyanova, M., Van Laer, P., Baig, M.Z., Casson, N., Marzooqi, H., Suwaidi, S.J., Ali, H., Subbiah, S.K., D'Mello, A., 2017. Shedding the first light on the Shilaif source rock in the syncline, onshore Abu Dhabi. Presented at ADIPEC, Abu Dhabi, UAE, 13–16 November. SPE-188762-MS. <https://doi.org/10.2118/188762-MS>
- EIA (U.S. Energy Information Administration). 2015. Technically recoverable shale oil and shale gas resources: United Arab Emirates. 30 pp.
- Eltom, H. and Hasiotis, S.T. 2019. Lateral and vertical trends of preferred flow pathways associated with bioturbated carbonate: Examples from Middle to Upper Jurassic strata, central Saudi Arabia. In: McNeill, D.F., Harris, P.M., Rankey, E.C., and Hsieh, J.C., eds., *Carbonate Pore Systems: New Developments and Case Studies (Mountjoy Symposium 2)*: SEPM Special Publication **112**: 126–140. <https://doi.org/10.2110/sepmssp.112.05>
- Eltom, H.A., Rankey, E.C., Hasiotis, S.T., and Barati, R. 2019. Effect of bioturbation on petrophysical properties: insights from geostatistical and flow simulation modeling: *Marine and Petroleum Geology* **104**: 259–269. <https://doi.org/10.1016/j.marpetgeo.2019.03.019>
- Forster, A., Schouten, S., Moriya, K., Wilson, P.A., and Sinninghe Damsté, J.S. 2007. Tropical warming and intermittent cooling during the Cenomanian-Turonian oceanic anoxic event 2: Sea surface temperature records from the equatorial Atlantic: *Paleoceanography and Paleoclimatology* **22** (1): 14 pp. <https://doi.org/10.1029/2006PA001349>
- Gramajo, E. and Rached, M.R. 2022. Not all unconventional reservoirs are similar MENA regional vs global anisotropic rock index and mechanical characterization – Part 2: ADIPEC, Abu Dhabi. Presented at ADIPEC, Abu Dhabi, UAE, 31 October–1 November. SPE-210990-MS. <https://doi.org/10.2118/210990-MS>
- Haq, B.U. 2014. Cretaceous eustasy revisited: *Global and Planetary Change* **113**: 44–58. <https://doi.org/10.1016/j.gloplacha.2013.12.007>
- Haq, B.U., Hardenbol, J. and Vail, P.R. 1987. Chronology of fluctuating sea levels since the Triassic: *Science* **235** (4793): 1156–1167. <https://doi.org/10.1126/science.235.4793.1156>

- Hunt, D., and Tucker, M.E. 1992. Stranded parasequences and the forced regressive wedge systems tract: deposition during base-level fall: *Sedimentary Geology* **81** (1-2): 1–9. [https://doi.org/10.1016/0037-0738\(92\)90052-S](https://doi.org/10.1016/0037-0738(92)90052-S)
- Jenkyns, H.C. 2010. Geochemistry of oceanic anoxic events: *Geochemistry, Geophysics, Geosystems* **11** (3): 30 pp. <https://doi.org/10.1029/2009GC002788>
- Jenkyns, H.C. 1980. Cretaceous anoxic events: from continents to oceans: *Journal of the Geological Society* **137**: 171–188. <https://doi.org/10.1144/gsjgs.137.2.017>
- Kietzmann, D.A., Palma, R.M., Riccardi, A.C., Martín-Chivelet, J., and López-Gómez, J., 2014. Sedimentology and sequence stratigraphy of a Tithonian–Valanginian carbonate ramp (Vaca Muerta Formation): A misunderstood exceptional source rock in the Southern Mendoza area of the Neuquén Basin, Argentina, Mendoza: *Sedimentary Geology* **302**: 64–86. <https://doi.org/10.1016/j.sedgeo.2014.01.002>
- MacEachern, J.A., Raychaudhuri, I., and Pemberton, S.G. 1992. Stratigraphic applications of the Glossifungites ichnofacies: Delineating discontinuities in the rock record. In: Pemberton, S.G., ed., Applications of Ichnology to Petroleum Exploration: SEPM, Core Workshop **17**: 169–198. <https://doi.org/10.2110/cor.92.01.0169>
- MacEachern, J.A., Bann, K.L., Gingras, M.K. and Pemberton, S.G. 2007. The ichnofacies paradigm: High-resolution paleoenvironmental interpretation of the rock record. In: MacEachern, J.A., Bann, K.L. Gingras, M.K., and Pemberton, S.G., eds., *Applied Ichnology: SEPM Short Course Notes* **52**, 27–64. <https://doi.org/10.2110/pec.07.52.0027>
- McMahon, T.P., Larson, T.E., Zhang, T., and Shuster, M. 2024. Geologic characteristics, exploration and production progress of shale oil and gas in the United States: An overview: *Petroleum Exploration and Development* **51** (4): 925–948. [https://doi.org/10.1016/S1876-3804\(24\)60516-1](https://doi.org/10.1016/S1876-3804(24)60516-1)
- Middleton, G.V. and Hampton, M. A. 1973. Sediment gravity flows: mechanics of flow and deposition. In: Middleton, G.V. and Bouma, A.H., eds., *Turbidites and Deep-Water Sedimentation: SEPM, Pacific Section*: 38 pp.
- Mort, H.P., Adatte, T., Keller, G., Bartels, D., Föllmi, K. B., Steinmann, P., Berner, Z., and Chellai, E.H. 2008. Organic carbon deposition and phosphorus accumulation during Oceanic Anoxic Event 2 in Tarfaya, Morocco: *Cretaceous Research* **29** (5-6): 1008–1023. <https://doi.org/10.1016/j.cretres.2008.05.026>
- O'Brien, C.L., Robinson, S.A., Pancost, R.D., Sinninghe Damsté, J. S., Schouten, S., Lunt, D. J., Alsenz, H., Bornemann, A., Bottini, C., Brassell, S.C., Farnsworth, A., Forster, A., Huber, B.T., Inglis, G.N., Jenkyns, H.C., Linnert, C., Littler, K., Markwick, P., McAnena, A., Mutterlose, J., Naafs, B. D.A., Püttmann, W., Sluijs, A., van Helmond, N.A.G.M., Vellekoop, J., Wagner, T., and Wrobel, N.E. 2017. Cretaceous sea-surface temperature evolution: Constraints from TEX<sub>86</sub> and planktonic foraminiferal oxygen isotopes: *Earth-Science Reviews* **172**: 224–247. <https://doi.org/10.1016/j.earscirev.2017.07.012>
- Pemberton, S.G. and Gingras, M.K. 2005. Classification and characterizations of biogenically enhanced permeability: *AAPG Bulletin* **89** (11): 1493–1517. <https://doi.org/10.1306/07050504121>
- Plint, A.G. (1988). Sharp-based shoreface sequences and 'offshore bars' in the Cardium Formation of Alberta: Their relationship to relative changes in sea level. In: Wilgus, C.K., Hastings, B.S., Posamentier, H.W., Van Wagoner, J.C., Ross, C.A., and Kendall, C.G.St.C., eds., *Sea-Level Changes: An Integrated Approach: SEPM, Special Publication* **42**, 357–370. <https://doi.org/10.2110/pec.88.01.0357>
- Savrdá, C.E. 1991. Ichnology in sequence stratigraphic studies: An example from the lower Palaeocene of Alabama: *Palaos* **6** (1): 39–53. <https://doi.org/10.2307/3514952>
- Schlanger, S.O. and Jenkyns, H.C. 1976. Cretaceous Oceanic Anoxic events: Causes and consequences: *Geologie en Mijnbouw* **55** (3-4): 179–184.
- Sharland, P.R., Archer, R., Casey, D.M., Davies, R.B., Hall, S.H., Heward, A.P., Horbury, A.D. and Simmons, M.D. 2001. Arabian plate sequence stratigraphy: *GeoArabia Special Publication 2, Gulf PetroLink, Bahrain*. 371 pp.
- Steiner, S., Ahsan, S.A., Noufal, A., Franco, B., Koksalan, T., Amjad, K., Helja, E., Alhosani, S., and Adesanya, A. 2015. Integrated approach to evaluate unconventional and tight reservoirs in Abu Dhabi. Presented at ADIPEC, Abu Dhabi, UAE, 9-12 November. SPE-177610-MS. <https://doi.org/10.2118/177610-MS>
- Taher, A.K. 2010. Unconventional oil exploration potential in Early Mature source rock Kitchens. Presented at ADIPEC, Abu Dhabi, UAE, 1-4 November. SPE-137897-MS. <https://doi.org/10.2118/137897-MS>
- Trabelsi, A. and Beg, M.A. 2000. Characterization and mapping of burrowed and microporous intervals in the Arab D Reservoir, Dukhan Field, Qatar. Presented at ADIPEC, Abu Dhabi, UAE, 13-15 October. SPE-87238-MS. <https://doi.org/10.2118/87238-MS>
- Trabucho Alexandre, J., van Gilst, R.I., Rodríguez-López, J.P., and de Boer, P.L. 2011. The sedimentary expression of oceanic anoxic event 1b in the North Atlantic: *Sedimentology* **58**: 1217–1246. <https://doi.org/10.1111/j.1365-3091.2010.01202.x>
- Turner, J.T., 2002. Zooplankton fecal pellets, marine snow and sinking phytoplankton blooms: *Aquatic Microbial Ecology* **27**: 57–102. <https://doi.org/10.3354/ame027057>
- Turner, J.T. 2015. Zooplankton fecal pellets, marine snow, phytodetritus and the ocean's biological pump: *Progress in Oceanography* **130**: 205–248. <https://doi.org/10.1016/j.pocean.2014.08.005>



- Tyson, R.V. 1995. Sedimentary Organic Matter: Organic Facies and Palynofacies. *Chapman and Hall, London*: 615 pp.
- Vahrenkamp, V., Franco, B.J., Popa, D., Barata, J., Grelaud, C., Razin, P., Grosheny, D., and Bulot, L. 2015. Development and infill of the Late Albian to Turonian Shilaif intrashelf basin at the eastern margin of the giant Mesozoic Arabian carbonate platform: Basin architecture and time stratigraphy. Presented at IPTC, Doha, Qatar, 6-9 December. IPTC-18488-MS. <https://doi.org/10.2523/IPTC-18488-MS>
- Van Laer, P., Leyrer, K., Povstyanova, M., Baig, M.Z., Makarychev, G., and Al-Marzooqi, H. 2019. Cenomanian Shilaif unconventional shale oil potential in onshore Abu Dhabi, UAE. Presented at URTEC, Denver, Colorado, USA, 22-24 July. URTEC-2019-526-MS. <https://doi.org/10.15530/urtec-2019-526>
- Wang, Y., Huang, C., Sun, B., Quan, C., Wu, J., and Lin, Z. 2014. Paleo-CO<sub>2</sub> variation trends and the Cretaceous greenhouse climate: *Earth-Science Reviews* **129**: 136–147. <https://doi.org/10.1016/j.earscirev.2013.11.001>
- Wilson, R.S., Chitale, J., Huffman, K., Montgomery, P., and Prochnow, S.J. 2020. Evaluating the depositional environment, lithofacies variation, and diagenetic processes of the Wolfcamp B and lower Spraberry intervals in the Midland Basin: Implications for reservoir quality and distribution: *AAPG Bulletin* **104** (6): 1287–1321. <https://doi.org/10.1306/12031917358>